Diseño de Modelos Analíticos en dbt a partir de Datos Ingestados con Airbyte

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Descripción

Pipeline ELT implementado con **Airbyte Cloud** (ingesta desde CSV públicos), **MotherDuck** (DuckDB en la nube como destino) y **dbt Core** (transformación en 3 capas). Repositorio en:

`https://github.com/Jessbpd/dbt-airbyte-proyecto`

1. Fuentes de Datos en Airbyte

Se configuraron dos sources en Airbyte Cloud usando el conector **File (CSV)** apuntando a archivos públicos en GitHub, con destino MotherDuck en el schema `airbyte_raw`.

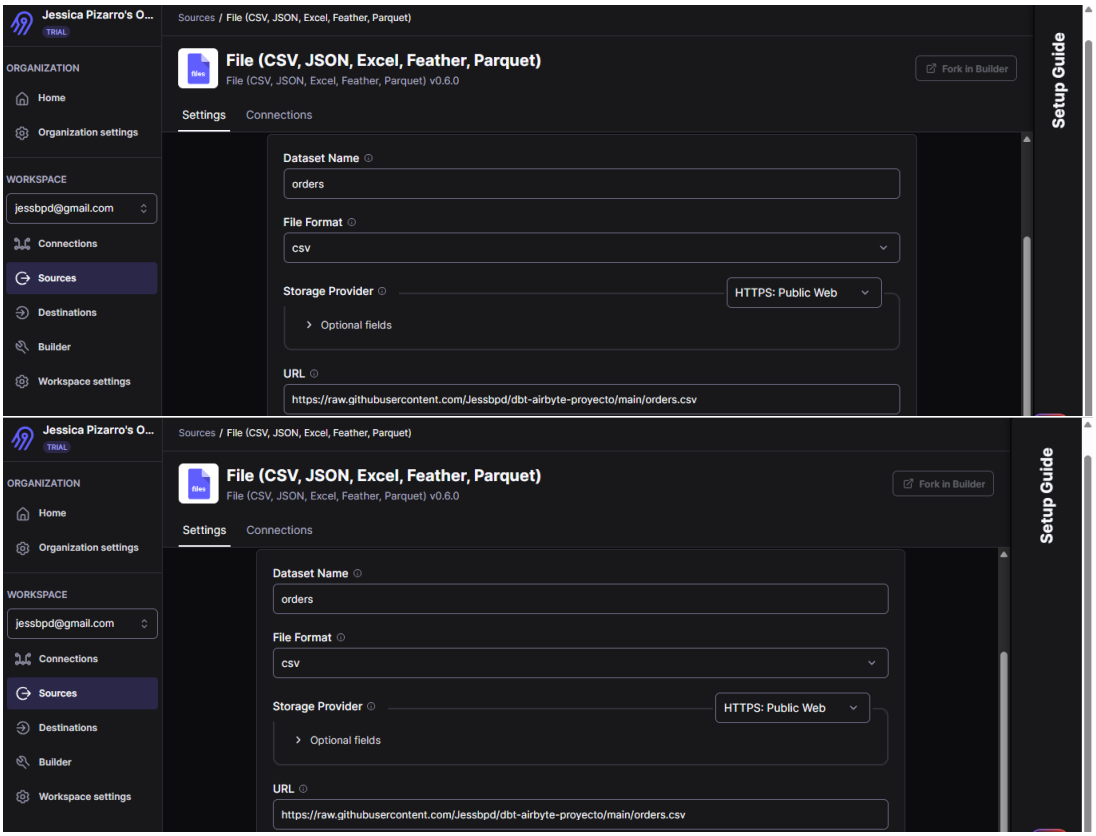

Figura 1: Sources `customers-csv` y `orders-csv` sincronizados en Airbyte Cloud

2. Proyecto dbt — Estructura y Modelos

El proyecto sigue la arquitectura de 3 capas estándar de dbt:

```
dbt_airbyte_proyecto/
          dbt_project.yml
          models/
              staging/
                    _sources.yml
                    stg_customers.sql
                    stg_orders.sql
              intermediate/
                    int_customer_orders.sql
              marts/
                  mart_sales_summary.sql
```

2.1. Archivo `_sources.yml`

Declara las dos tablas cargadas por Airbyte con tests de calidad (`unique`, `not_null`) en las columnas clave.

2.2. Staging

Dos modelos que limpian y estandarizan los datos crudos de Airbyte: `stg_customers` normaliza emails y fechas; `stg_orders` convierte tipos y calcula `total_amount`.

2.3. Intermediate

`int_customer_orders` une órdenes con clientes mediante `LEFT JOIN` en `customer_id`, enriqueciendo cada orden con nombre, país y plan del cliente.

2.4. Mart

`mart_sales_summary` es una OBT (One Big Table) materializada como tabla física, con métricas agregadas por fecha, cliente y producto: `total_orders`, `total_revenue` y `avg_order_value`.

3. Ejecución y Resultados

```
C:\WINDOWS\system32\cmd.

C:\Users\USUARIO\Documents\dbt_airbyte_proyecto>dbt run
19:08:27  Running with dbt=1.11.7
19:08:27  Registered adapter: duckdb=1.10.1
19:08:28  Unable to do partial parsing because saved manifest not found. Starting full parse.
19:08:29  Found 4 models, 12 data tests, 2 sources, 474 macros
19:08:29
19:08:29  Concurrency: 4 threads (target='dev')
19:08:29
19:08:33  1 of 4 START sql view model airbyte_raw_staging.stg_customers ................... [RUN]
19:08:33  2 of 4 START sql view model airbyte_raw_staging.stg_orders ...................... [RUN]
19:08:34  1 of 4 OK created sql view model airbyte_raw_staging.stg_customers ............. [OK in 1.04s]
19:08:34  2 of 4 OK created sql view model airbyte_raw_staging.stg_orders ................ [OK in 1.04s]
19:08:34  3 of 4 START sql view model airbyte_raw_intermediate.int_customer_orders ....... [RUN]
19:08:35  3 of 4 OK created sql view model airbyte_raw_intermediate.int_customer_orders .. [OK in 0.99s]
19:08:35  4 of 4 START sql table model airbyte_raw_marts.mart_sales_summary ............. [RUN]
19:08:36  4 of 4 OK created sql table model airbyte_raw_marts.mart_sales_summary ......... [OK in 1.19s]
19:08:36
19:08:36  Finished running 1 table model, 3 view models in 0 hours 0 minutes and 7.04 seconds (7.04s).
19:08:36
19:08:36  Completed successfully
19:08:36
19:08:36  Done. PASS=4 WARN=0 ERROR=0 SKIP=0 NO-OP=0 TOTAL=4
```

Figura 2: `dbt run`: 4 modelos ejecutados exitosamente (PASS=4)

```
C:\WINDOWS\system32\cmd.

C:\Users\USUARIO\Documents\dbt_airbyte_proyecto>dbt test
19:09:03  Running with dbt=1.11.7
19:09:03  Registered adapter: duckdb=1.10.1
19:09:04  Found 4 models, 12 data tests, 2 sources, 474 macros
19:09:04
19:09:04  Concurrency: 4 threads (target='dev')
19:09:04
19:09:07  1 of 12 START test not_null_stg_customers_customer_id ........................ [RUN]
19:09:07  3 of 12 START test not_null_stg_orders_customer_id ........................... [RUN]
19:09:07  2 of 12 START test not_null_stg_customers_email ............................... [RUN]
19:09:07  4 of 12 START test not_null_stg_orders_order_id ............................... [RUN]
19:09:07  2 of 12 PASS not_null_stg_customers_email ..................................... [PASS in 0.28s]
19:09:07  4 of 12 PASS not_null_stg_orders_order_id ..................................... [PASS in 0.27s]
19:09:07  3 of 12 PASS not_null_stg_orders_customer_id ................................... [PASS in 0.29s]
19:09:07  1 of 12 PASS not_null_stg_customers_customer_id ............................... [PASS in 0.29s]
19:09:07  5 of 12 START test source_not_null_customers_source_customers_customer_id ...... [RUN]
19:09:07  6 of 12 START test source_not_null_customers_source_customers_email ........... [RUN]
19:09:07  7 of 12 START test source_not_null_orders_source_orders_customer_id ........... [RUN]
19:09:07  8 of 12 START test source_not_null_orders_source_orders_order_id ............... [RUN]
19:09:08  5 of 12 PASS source_not_null_customers_source_customers_customer_id ........... [PASS in 0.27s]
19:09:08  7 of 12 PASS source_not_null_orders_source_orders_customer_id ................. [PASS in 0.26s]
19:09:08  6 of 12 PASS source_not_null_customers_source_customers_email ................. [PASS in 0.27s]
19:09:08  9 of 12 START test source_unique_customers_source_customers_customer_id ........ [RUN]
19:09:08  10 of 12 START test source_unique_orders_source_orders_order_id ............... [RUN]
19:09:08  8 of 12 PASS source_not_null_orders_source_orders_order_id ..................... [PASS in 0.26s]
19:09:08  11 of 12 START test unique_stg_customers_customer_id ........................... [RUN]
19:09:08  12 of 12 START test unique_stg_orders_order_id ................................ [RUN]
19:09:08  9 of 12 PASS source_unique_customers_source_customers_customer_id ............. [PASS in 0.27s]
19:09:08  10 of 12 PASS source_unique_orders_source_orders_order_id ...................... [PASS in 0.28s]
19:09:08  11 of 12 PASS unique_stg_customers_customer_id ................................ [PASS in 0.28s]
19:09:08  12 of 12 PASS unique_stg_orders_order_id ..................................... [PASS in 0.26s]
19:09:08
19:09:08  Finished running 12 data tests in 0 hours 0 minutes and 4.00 seconds (4.00s).
19:09:08
19:09:08  Completed successfully
19:09:08
19:09:08  Done. PASS=12 WARN=0 ERROR=0 SKIP=0 NO-OP=0 TOTAL=12
```

Figura 3: `dbt test`: 12 tests de calidad pasados (PASS=12)

4. DAG — Linaje de Datos

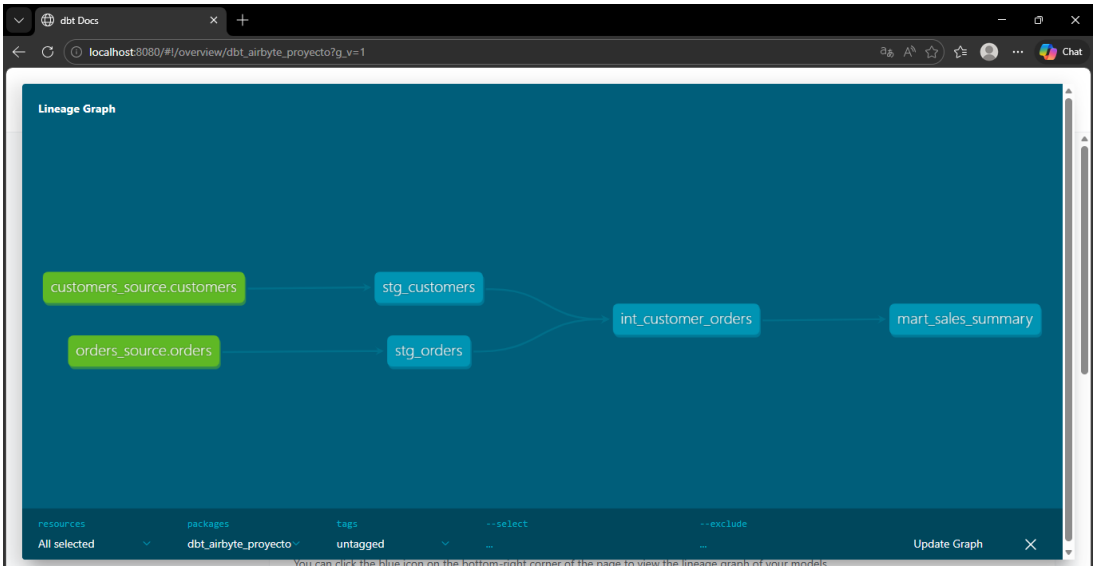

Figura 4: DAG generado por `dbt docs`: 2 sources → 2 staging → 1 intermediate → 1 mart